Technical Service Bulletin

Technical Service Bulletin: tsb100441	Released Date: 26-Jul-2006
QST30 Tier 2 Engine Introduction	

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Note : This Technical Service Bulletin supersedes and replaces Service/Parts Topics 97t0-7 and 99t0-22.

This Technical Service Bulletin announces the introduction of the QST30 Tier 2, Tier 1, non-certified products, and identifies unique engine components for the non-certified products.

The QST30 Tier 2 engine is redesigned to address the US Environmental Protection Agency (EPA) exhaust emissions levels for off-highway engines in the >560 kW [>751 HP] power category built after 01 January 2006. It is built to meet and exceed present and future competitive pressures, worldwide emissions regulations, and corporate standards for performance, reliability and durability.

QST30 Tier 2 Industrial		
Configuration Number	D573002CX03	D573003CX03
Number of Cylinders	12	12
Displacement	30.5 liters [1860 in ^{3}]	30.5 liters [1861 in ^{3}]
Bore	140 mm [5.51 in]	140 mm [5.51 in]
Stroke	165 mm [6.5 in]	165 mm [6.5 in]
Compression Ratio	15.7:1	14.7:1
Aspiration	Turbocharged and Charge Air Cooled	Turbocharged and 2-Pump 2-Loop Aftercooled
Approximate Engine Weight	3328 kg [7337 lbs]	3328 kg [7337 lbs]
Maximum Overspeed Capacity	2625 rpm	2625 rpm

QST30 Tier 2 Power Generation	
Configuration Number	D573001GX03

QST30 Tier 2 Power Generation	
Number of Cylinders	12
Displacement	30.5 liters [1860 in ^{3}]
Bore	140 mm [5.51 in]
Stroke	165 mm [6.5 in]
Compression Ratio	14.7:1
Aspiration	Turbocharged and Charge Air Cooled, 2-Pump 2-Loop Aftercooled, Jacket Water Aftercooled
Approximate Engine Weight	3112 kg [6860 lbs]
Maximum Overspeed Capacity	2625 rpm

QST30 Tier 2 Engine Ratings (Industrial)

The fuel ratings available for the QST30 Tier 2 engine are essentially a carryover from the QST30 Tier 1 engine. The performance parts have been developed to meet Tier 2 emission levels, achieve the best possible fuel consumption, and achieve similar transient response as the Tier 1 engine.

The Tier 2 ratings shown below are planned for production availability. Additional ratings, performance curves, and data sheets are available and can be obtained at a Cummins Authorized Service Location.

QST30 Configuration Number - D573002CX03				
Option Number	**CPL Number**	**Power kW [bhp] @ rpm**	**Peak Torque N•m [ft-lb] @ rpm**	**Rating Type**
FR 5234	1244	895 [1200] @ 2100	5084 [3750] @ 1400	Restricted
FR 5235	1244	895 [1200] @ 1900	5085 [3751] @ 1400	Restricted
FR 5236	1244	820 [1100] @ 2100	4629 [3414] @ 1400	Intermittent
FR 5237	1244	783 [1050] @ 2100	4629 [3414] @ 1300	Intermittent
FR 5239	1244	783 [1050] @ 1900	4629 [3414] @ 1300	Intermittent
FR 5238	1244	783 [1050] @ 1800	4629 [3414] @ 1300	Intermittent
FR 5240	1244	746 [1000] @ 2100	4629 [3414] @ 1300	Intermittent
FR 5241	1244	746 [1000] @ 1800	4629 [3414] @ 1300	Intermittent
FR 5242	1244	708 [950] @ 2100	4188 [3089] @ 1300	Intermittent

QST30 Configuration Number - D573002CX03				
Option Number	**CPL Number**	**Power kW [bhp] @ rpm**	**Peak Torque N•m [ft-lb] @ rpm**	**Rating Type**
FR 5243	1244	671 [900] @ 2100	3747 [2764] @ 1300	Intermittent
FR 5244	1244	634 [850] @ 2000	3806 [2807] @ 1400	Intermittent
FR 5245	1244	634 [850] @ 1800	3806 [2807] @ 1400	Continuous
FR 5246	1244	567 [760] @ 2100	3350 [2471] @ 1300	Intermittent

Engine Ratings (Power Generation)

The fuel ratings available for the QST30 Tier 2 engine are essentially a carryover from the QST30 Tier 1 engine. The performance parts have been developed to meet Tier 2 emission levels, achieve the best possible fuel consumption, and achieve similar transient response as the Tier 1 engine. The QST30 Tier 2 ratings shown in the table below are planned for production availability. Additional ratings can be requested by submitting a Value Package Change Request (VPCR). Additional ratings, performance curves, and data sheets are available and can be obtained at a Cummins Authorized Service Location.

		Standby		**Prime**	**Continuous**	**Cooling**		
		1500	**1800**	**1500**	**1800**		**1500**	**1800**
		HP/KWm	**HP/KWm**	**HP/KWm**	**HP/KWm**		**HP/KWm**	**HP/KWm**
FR 5247	QST30-G5 NR2	N/A	1490/1112	N/A	1350/1007	N/A	1115/832	2P2L
FR 5250	QST30-G5 NR2	N/A	1490/1112	N/A	1350/1007	N/A	1115/832	ATA

		Standby		Prime	Continuous	Cooling		
		1500	1800	1500	1800		1500	1800
		HP/KWm	HP/KWm	HP/KWm	HP/KWm		HP/KWm	HP/KWm
Additional ratings, performance curves, and data sheets are available and can be obtained at a Cummins Authorized Service Location.								

The QST30 Industrial Tier 1 service parts topic introduced the Tier 1 emission-certified QST30 engine for mobile off-highway applications. A new procedure for the charge-air cooler (CAC), a new symptom tree, and revisions to five other symptom trees were also included in the Tier 1 engine introduction.

To meet year 2000 Environmental Protection Agency (EPA) and California Air Resources Board (CARB) Tier 1 emission levels, it was necessary to change the non-certified QST30 aspiration from turbocharged and jacket water aftercooled (JWAC) to turbocharged and charge-air cooled (CAC). The Tier 1 product employs two separate cooling circuits; A liquid-coolant-filled high-temperature circuit (HTC) for the main engine jacket water cooling, and a low-temperature, air-to-air circuit (LTC) for charge-air cooling. This topic addressed **only** the differences between the Tier 1 and non-certified QST30 Industrial engines.

Charge-air cooling of the intake air was required to retain engine efficiency while injection timing was retarded to obtain lower NOx. Charge-air cooling also allowed higher horsepower capabilities for the equivalent thermal loading. In addition to the non-certified QST30 power ranges of 760 hp to 1050 hp, a 1200 hp with 6000-foot altitude capability became available. The new offering also featured a 25 percent torque rise and is restricted to intermittent applications with load factors equal to or less than 40 percent.

QST30 Industrial Tier 1 limited production began 01 January 2000 and retained the configuration number D573002CX03.

For general charge-air cooling installation guidelines, see the Application Engineering Bulletin Number 3884966, (AEB) 24.06 - 1999 Industrial Application Installation Recommendations, Charge-Air Cooling.

Emission Regulations

Year 2000 Tier 1 Emission Regulations		
Emission Regulating Agency	**Emission Limit (g/bhp-hr) NOx**	**Horsepower**
CARB	6.9 NOx	Above 750 hp
EPA	6.9 NOx	Above 750 hp

Ratings

Note : The only difference between CPL 2640 and CPL 2818 is the orientation of the turbocharger compressor housings. The base turbocharger part number is the same for the two CPLs.

QST30 Industrial Tier 1 Ratings				
Engine Model	**Rated Horsepower kW [bhp] rpm, N•m [ft-lb] rpm**	**Application**	**CPL**	**ECM Code**
QST30-C	669 [897]@2000 rpm. Torque peak: 4103 [3026]@1300 rpm	Komatsu WA900, front-out exhaust	2640	K50017
QST30-C	807 [897]@2000 rpm. Torque peak: 4629 [3414]@1400 rpm	Komatsu 330M, rear-out exhaust, with PMC, with CENSE™	2818	K50019
QST30-C	567 [760]@2100 rpm. Torque peak: 3350 [2471]@1300 rpm	Kawasaki 135Z	2640	K50018

QST30 Industrial Tier 1 Ratings				
Engine Model	**Rated Horsepower kW [bhp] rpm, N•m [ft-lb] rpm**	**Application**	**CPL**	**ECM Code**
Additional ratings, performance curves, and data sheets are available and can be obtained at a Cummins Authorized Service Location.				

The QST30-C Industrial service parts topic announced the introduction of the QST30-C Industrial engine.

New industrial ratings of the QST30 engine were released for limited production in September, 1997. Four ratings of this industrial engine were released. The QST30-C uses many of the successfully proven technologies already in production on QST30 Power Generation engines. The QST30-C uses a full-authority electronic Bosch® fuel system, Cummins Inc. developed controls, and is fully compatible with the CENSE™ engine monitoring system. QST30-C Industrial ratings are designed to replace V28 and lower K38 ratings with improved fuel economy, transient response, emissions, and diagnostic capabilities.

QST30-C Non-Certified Industrial	
Model Name	QST30-1050
Configuration Number	D573002CX03
CPL	2127
Fuel Code	105
Injector (Qty.)	3092022 (2), 3092023 (10)
Crankshaft	3092922
Camshft (Qty.)	3092971 (1), 3092974 (1).
Piston (Qty.)	3092606 (6), 3092607, (6).
Turbocharger (Qty.)	3537114 (1), 3097115 (1).
Cylinder Head	3094190
Horsepower	1050 hp @ 2100 rpm

QST30-C Non-Certified Industrial	
Torque	3415 ft-lb @ 1300 rpm

The QST30-C Industrial engine contains several unique parts. A general description of these components and a listing of parts that are not shared with the power generation QST30 engines. Components are organized by group for ease of use.

Note : The QST30-C engines used in Komatsu applications will be Komatsu branded and designated SA12V140 Z-1. They will not be Cummins Inc. labeled or identified as QST30-C. The engines in Komatsu applications are not covered by Cummins Inc. warranty, and they will be serviced through the Komatsu service network.

Last Modified: 26-Jul-2006